PROJECT PROPOSAL

Autonomous Elderly Care Monitoring System

Raspberry Pi Competition 2026 · Entry Submission

RASPBERRY PI COMPETITION 2026

Autonomous Monitoring · Assistive Robotics
Elderly Care Technology

VISION ACTIVE

IRIS

INTELLIGENT RECOGNITION & INCIDENT SURVEILLANCE

A **robot that watches over elderly care home residents** so staff don't have to be everywhere at once. It patrols autonomously, spots when someone isn't moving, and raises the alarm - all without anyone needing to do a thing.

HOW IT WORKS

01
Patrol
Moves around the room on its own - no remote control needed

02
Stream
Pi Camera sends live frames over a Python socket connection

03
Analyse
MediaPipe tracks the nose-tip - stillness for too long means trouble

04
Alert
Staff get an instant alert to check on the resident straight away

SYSTEM ARCHITECTURE

Pi Camera captures frames → raw frames → **Raspberry Pi** encodes & sends → Python socket → **Laptop** runs pipeline → landmarks → **MediaPipe** stillness check → PWM signal → **Servos** drive wheels

Hardware Stack

COMPUTE	Raspberry Pi 4B
VISION	Pi Camera Module V2
MOTION	2x SG90-HV Servos
POWER	HW-131 Module - USB
COMMS	Python Sockets / WiFi

Estimated Cost

Raspberry Pi 4B	·£50
Pi Camera V2	·£20
2x SG90-HV Servos	·£10
HW-131 + misc	·£10
TOTAL	·£90

What makes it special

- No wearables - residents don't have to do anything
- Nose-tip tracking catches stillness others miss
- Pi handles everything - no extra microcontroller needed
- Runs 24/7 without any staff input
- Built on affordable, off-the-shelf parts
- Easy to extend with extra sensors or alerts

At a glance

24/7
Always watching

0
Wearables needed

<10s
Stillness to alert

WHY IT MATTERS

1 in 3
adults over 65 fall at least once a year, often going unnoticed for hours

500k+
care home residents in the UK with no continuous automated monitoring

minutes
can be the difference between full recovery and serious long-term harm